

»» Chapter XV ««

MÖBIUS TRANSFORMATIONS

PREFATORY REMARKS

The study of the univalent meromorphic functions with domain $\hat{\mathbb{C}}$, the Möbius transformations, was begun in Ex. 4.25, Ch. VI. These functions have long played an important part in the geometric theory of functions. Their study is a preliminary to the theory of automorphic groups and functions, a subject initiated by Klein and Poincaré and still provoking lively mathematical interest. Hence for the student wishing to pursue modern complex function theory further, especially in the two areas mentioned, the basic facts about Möbius transformations are an essential part of his equipment. We shall encounter these transformations in the study of the elliptic modular function (Ch. XVI).

Our principal concern in the present chapter will be with the classification of Möbius transformations.

1. CLASSIFICATION OF MÖBIUS TRANSFORMATIONS

We have already noted (Ex. 4.25, Ch. VI) that if a and b are distinct points of $\hat{\mathbb{C}}$, there exists a Möbius transformation f satisfying $f(a) = 0$, $f(b) = \infty$. Thus $(z - a)/(z - b)$ serves when $a, b \in \mathbb{C}$; $(z - b)^{-1}$ when $a = \infty$; and $z - a$ when $b = \infty$. To what extent f is determined is easy to see. If g is a Möbius transformation also satisfying the same conditions as f , $g \circ f^{-1}(z) = cz$, $c \in \mathbb{C} - \{0\}$, and hence $g = cf$. The converse is obvious.

It will be convenient to let I denote the identity map of $\hat{\mathbb{C}}$ onto itself.

An equivalence relation. Given Möbius transformations L and M , we say that M is *equivalent* to L provided that there exists a Möbius transformation S such that

$$M = S \circ L \circ S^{-1}. \quad (1.1)$$

It is immediate that a bona fide equivalence relation is defined thereby. To check this, we note (1) when $L = M$, I serves for S , (2) from (1.1) we have $L = S^{-1} \circ M \circ (S^{-1})^{-1}$, (3) if N and T are Möbius transformations satisfying $N = T \circ M \circ T^{-1}$, then $N = (T \circ S) \circ L \circ (T \circ S)^{-1}$. Clearly, I is equivalent only to itself.

Let M be a Möbius transformation, $\neq I$. We shall see that the number of *fixed points* of M (that is, the $z \in \hat{\mathbb{C}}$ satisfying $M(z) = z$) is either one or two. Suppose that $M(\infty) = \infty$, so that $M(z) = az + b$, $a \neq 0$. If $a = 1$, then $b \neq 0$ and ∞ is the only fixed point. If $a \neq 1$, then the fixed points of M are ∞ and $(1 - a)^{-1}b$. Suppose that $M(\infty) \neq \infty$. Then $M - I$ has constant valence two and has ∞ as a fixed point. The fixed points of M are the zeros of $M - I$ and are at most two in number.

We continue and consider the case where M has *exactly one* fixed point, say a . Let S denote a Möbius transformation satisfying $S(a) = \infty$.† Now $S \circ M \circ S^{-1}$ has just one fixed point, ∞ . Hence $S \circ M \circ S^{-1}(z) = z + b$, $b \neq 0$. Using $b^{-1}[(bz) + b] = z + 1$, we conclude: *all Möbius transformations with just one fixed point are mutually equivalent*. In particular, they are all equivalent to the translation through 1 (extended to $\hat{\mathbb{C}}$ in the obvious way). Möbius transformations with just one fixed point are termed *parabolic*. We conclude using the definition of equivalence that the set of parabolic transformations constitutes an equivalence class.

There remains to consider the case where M has *two distinct* fixed points, say a and b . Let S denote a Möbius transformation satisfying $S(a) = 0$, $S(b) = \infty$. Then $S \circ M \circ S^{-1} = cI$, $c \neq 0, 1$. When are two Möbius transformations of this special form equivalent? If $d \neq 0, 1$ and T is a Möbius transformation satisfying $dI = T \circ (cI) \circ T^{-1}$, we see that $T(0)$ and $T(\infty)$ are the fixed points of dI . Hence either $T = \alpha I$ or $T = \alpha(1/I)$ for some $\alpha \neq 0$. If the first possibility occurs, $d = c$; if the second occurs, $d = 1/c$. It is immediate that if $d = c$, or $d = 1/c$, such a T exists. We conclude that dI is equivalent to cI if and only if

$$c + c^{-1} = d + d^{-1}.$$

Given M with exactly two distinct fixed points, we introduce $\kappa(M)$ as the common value of $c + c^{-1}$ such that M is equivalent to cI . If N is also a Möbius transformation with exactly two distinct fixed points, then N is equivalent

† If $a = \infty$, we may take I ; if $a \neq \infty$, $1/(z - a)$.

to M if and only if $\kappa(N) = \kappa(M)$. If ∞ is a fixed point of M , so that $M(z) = cz + d$, $c \neq 0, 1$, then M is equivalent to cI and hence $\kappa(M) = c + c^{-1}$. If ∞ is not a fixed point of M , we see that $S \circ M = cS$ and $T \circ M = c^{-1}T$, $c \neq 0, 1$, where S is as in the preceding paragraph with a and b the fixed points of M and $T = 1/S$. From the first equality we conclude $M'(a) = c$ and from the second $M'(b) = c^{-1}$. Hence $\kappa(M) = M'(a) + M'(b)$.

The Möbius transformations with exactly two distinct fixed points are classified by the invariant $\kappa(M)$. The set of values taken by $\kappa(M)$ consists precisely of $\mathbb{C} - \{2\}$. We term M *elliptic* provided that $\kappa(M) \in [-2, 2]$, *hyperbolic* provided that $\kappa(M) \in]2, +\infty[$, and *loxodromic* otherwise.

We may sum up as follows:

Theorem 1.1: *The equivalence classes of the set of Möbius transformations relative to the stated equivalence are: $\{I\}$; the set of parabolic transformations; and the sets C_a , $a \in \mathbb{C} - \{2\}$, where C_a is the set of Möbius transformations M having exactly two distinct fixed points and satisfying $\kappa(M) = a$.*

EXERCISES

1.1. Given $a, b, c \in \hat{\mathbb{C}}$ distinct, show that there exists a unique Möbius transformation M satisfying $M(a) = 0$, $M(b) = \infty$, $M(c) = 1$. Given (a, b, c, d) and $(\alpha, \beta, \gamma, \delta)$ where the components of each ordered quadruple are distinct members of $\hat{\mathbb{C}}$, show that a necessary and sufficient condition that there exist a Möbius transformation T satisfying $T(a) = \alpha$, $T(b) = \beta$, $T(c) = \gamma$, $T(d) = \delta$, is that $M(d) = L(\delta)$, where L is the unique Möbius transformation satisfying $L(\alpha) = 0$, $L(\beta) = \infty$, $L(\gamma) = 1$. Formulate a notion of equivalence for ordered triples of distinct points of $\hat{\mathbb{C}}$ relative to the group of Möbius transformations (as in Ex. 3.5, Ch. X). Show that all such ordered triples are equivalent to one another. Carry out the same program for ordered quadruples of distinct points of $\mathbb{C}$, determining the equivalence classes and a complete invariant.

1.2. For what c is cI elliptic, hyperbolic?

1.3. Suppose that the Möbius transformation $M \neq I$ does not have ∞ as a fixed point. Show that if M is parabolic and a is its fixed point, then $M'(a) = 1$. Show that if M has two distinct fixed points a and b , then $M'(a)M'(b) = 1$. This exercise is to be done without appeal to the auxiliary S by considering the poles of $1/(I - M)$.

1.4. We have seen in Ex. 4.25, Ch. VI, that a Möbius transformation M admits a representation on $\mathbb{C}$ of the form

$$M(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0.$$

Show that $(\alpha z + \beta)/(\gamma z + \delta)$ is also such a representation if and only if there exists $\lambda \in \mathbb{C} - \{0\}$ such that $\alpha = \lambda a$, $\beta = \lambda b$, $\gamma = \lambda c$, $\delta = \lambda d$. Consider the determination of the fixed points of M in terms of a, b, c, d and in the case of exactly two distinct fixed points the determination of $\kappa(M)$. Consider the relation of a representation of $L \circ M$ to representations of L and M , where L is also a Möbius transformation.

1.5. A Möbius transformation M maps $\mathbf{R} \cup \{\infty\}$ onto itself if and only if it admits a representation of the form

$$M(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbf{R}, \quad |ad - bc| = 1.$$

Show that M maps $\{\operatorname{Im} z > 0\}$ onto itself if and only if M admits such a representation with $a, b, c, d \in \mathbf{R}$, $ad - bc = 1$.

1.6. The set of Möbius transformations mapping $\{\operatorname{Im} z > 0\}$ onto itself for which a, b, c, d of a representation of the form given in Ex. 1.5 are integers constitutes a group with respect to composition. [This is the *modular group* to be considered in Ch. XVI.]

1.7. Given $f \in \mathbf{M}(\hat{\mathbf{C}})$ not constant. Show that if $g \in \mathbf{M}(\hat{\mathbf{C}})$ satisfies $f \circ g = f$, then g is a Möbius transformation. Show also that the set of such g constitutes a finite group G relative to the operation of composition. Show that if a is a constant whose value is not a fixed point of any $g \in G$ other than I and

$$h = \sum_{g \in G} \frac{1}{g - a},$$

then $f = F \circ h$ for a unique $F \in \mathbf{M}(\hat{\mathbf{C}})$.

Show for a finite group G of Möbius transformations that h as above is *automorphic* with respect to G , that is, $h \circ g = h$, $g \in G$, and that if $f \in \mathbf{M}(\hat{\mathbf{C}})$ is automorphic with respect to G , then as above $f = F \circ h$ for a unique $F \in \mathbf{M}(\hat{\mathbf{C}})$. Show furthermore that if $h_1 \in \mathbf{M}(\hat{\mathbf{C}})$ is automorphic with respect to G and also shares with h the property stated in the preceding sentence, then $h_1 = M \circ h$ where M is a Möbius transformation.